

# Computer Vision

*From Foundations to Frontier*

A 20-Chapter Modern Book — Outline & Writing Plan

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Target length: ~60,000 words / ~280 pages / ~3,000 words per chapter

Companion: GitHub repo with one runnable Colab notebook + production case study per chapter

Stack: Python 3.12, PyTorch 2.x, Hugging Face, Ultralytics, OpenCV, ONNX/TensorRT

Practitioner-first. Theory deep enough to debug. Every chapter ends in working code.

# Why this book, why now

By 2026 the CV landscape has structurally shifted three ways most existing books miss:

- **Frozen foundation backbones** (DINOv3, SAM 3) replace task-specific training for most use cases.
- **Vision-Language-Action models** (RT-2, OpenVLA,  $\pi$ 0, Helix) collapse perception + reasoning + control into one architecture.
- **Visual agents** (Claude Computer Use, GPT-4o vision agents) make vision the input layer of autonomous workflows.

This book covers the full arc — classical → CNN → ViT → foundation models → multimodal → embodied — with code that runs on a 2026 laptop and ships to production.

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# Part I — Foundations

## Chapter 1

### The 2026 Computer Vision Landscape

A grounded tour of where CV is in 2026: from the OpenCV era to foundation models. Maps the pipeline (acquisition → preprocessing → modeling → deployment), introduces the four eras (classical, CNN, ViT, foundation/multimodal), and frames the rest of the book. Reader leaves with a mental model of when to reach for SIFT vs YOLO vs SAM 3 vs a VLM, and sets up the running case-study dataset used through the book.

## Chapter 2

### Image Fundamentals and Pixel-Level Operations

Pixels, color spaces (RGB/HSV/LAB/YCbCr), sampling, quantization, histograms, and convolution from first principles. Covers filtering, blurring, sharpening, denoising (Gaussian, bilateral, non-local means), edge detection (Sobel, Canny, Laplacian), thresholding (Otsu, adaptive), and morphology. Hands-on: build a document-cleanup preprocessor in pure OpenCV that survives uneven lighting.

## Chapter 3

### Classical Feature Engineering: SIFT, ORB, and Geometry

Why classical features still matter in 2026 (SLAM, panorama, low-data domains). Covers Harris/FAST corners, SIFT/SURF/ORB/BRIEF descriptors, FLANN matching, RANSAC homography, Hough transforms, contour analysis, and template matching. Hands-on: stitch a multi-image panorama and build a logo-detector with feature matching — no neural networks.

## Chapter 4

### Camera Geometry, Calibration, and Stereo Vision

Pinhole and fisheye camera models, intrinsics/extrinsics, lens distortion, calibration with checkerboards, epipolar geometry, stereo rectification, disparity maps, and the math of triangulation. Bridges classical CV to modern depth estimation. Hands-on: calibrate a webcam and reconstruct a scene's depth map from a stereo pair using OpenCV's StereoSGBM.

# Part II — Deep Learning for Vision

## Chapter 5

### Convolutional Neural Networks: From LeNet to ConvNeXt

The CNN family tree as a single story: LeNet → AlexNet → VGG → Inception → ResNet → DenseNet → EfficientNet → MobileNet → ConvNeXt. Builds intuition for why each design choice (skip connections, bottleneck blocks, depthwise separables, modern normalizations) was a real fix to a real problem. Hands-on: implement a ResNet-18 from scratch in PyTorch and reproduce a CIFAR-10 result.

## Chapter 6

### Training Modern Vision Models in Practice

The unglamorous-but-decisive chapter: data loaders, Albumentations augmentation pipelines, mixup/cutmix, label smoothing, optimizers (SGD vs AdamW vs Lion), schedulers (cosine, one-cycle), mixed precision, gradient checkpointing, multi-GPU with PyTorch Lightning, and experiment tracking with Weights & Biases. Hands-on: fine-tune a timm model on a custom dataset and beat a published baseline through training tricks alone.

## Chapter 7

### Object Detection: From R-CNN to YOLO26 and DETR

Two-stage (R-CNN, Fast/Faster R-CNN) vs single-stage (SSD, RetinaNet, focal loss) vs anchor-free (FCOS, CenterNet) vs transformer-based (DETR, DINO-DETR, RT-DETR) vs the YOLO family through YOLOv8/YOLO11/YOLOv12/YOLO26 (NMS-free, end-to-end, edge-friendly). Covers mAP/IoU/PR-curve evaluation honestly. Hands-on: train YOLO26 on a custom Roboflow dataset and benchmark it against RT-DETR on the same task.

## Chapter 8

### Image Segmentation: U-Net to SAM 3 and Grounding DINO

Semantic vs instance vs panoptic segmentation. Walks through FCN, U-Net, DeepLab (atrous + ASPP), Mask R-CNN, Mask2Former, then jumps to the foundation-model era: SAM 1/2/3 (clicks, boxes, masks, and concepts via text in SAM 3), Grounding DINO + SAM for open-vocabulary segmentation. Hands-on: build a zero-shot 'segment anything I describe' pipeline with SAM 3 +

Grounding DINO and deploy it as a Gradio app.

# Part III — Modern Architectures and Foundation Models

## Chapter 9

### Vision Transformers and Hybrid Architectures

Attention from scratch, then ViT (patches, positional encoding, [CLS] token), DeiT (data-efficient training), Swin (hierarchical + shifted windows), MaxViT, and the new Mamba/SSM-based vision models (VMamba, Vision-Mamba). Discusses when transformers beat CNNs (data-rich, large-context) and when they lose (small-data, latency-sensitive). Hands-on: train a ViT-Small on ImageNette and compare its attention maps to a ResNet-50's class activation maps.

## Chapter 10

### Self-Supervised Vision: From SimCLR to DINOv3

Why labels became the bottleneck and how self-supervision broke it. Covers contrastive (SimCLR, MoCo, BYOL), masked image modeling (MAE, SimMIM), and the DINO line (DINO → DINOv2 → DINOv3, the 2025 model where a single frozen backbone outperforms specialists on dense prediction). Hands-on: extract DINOv3 features and use them as a frozen backbone for a downstream segmentation task with a 90% data reduction vs supervised baseline.

## Chapter 11

### CLIP and Vision-Language Pretraining

Contrastive language-image pretraining as a paradigm shift. Walks through CLIP, OpenCLIP, SigLIP, EVA-CLIP, and the open-vocabulary tasks they unlock: zero-shot classification, image-text retrieval, semantic search, and grounded detection. Covers prompt engineering for vision (templates, ensembles) and FAISS for billion-scale retrieval. Hands-on: build a 'search your photo library by natural language' app with CLIP + FAISS that runs on a laptop.

## Chapter 12

### Generative Vision: GANs, VAEs, and the Diffusion Era

GAN dynamics (mode collapse, Wasserstein loss), the VAE latent space, then the diffusion takeover: DDPM, score-based models, classifier-free guidance, latent diffusion, Stable Diffusion 3.5, Flux, SDXL, ControlNet, IP-Adapter, and LoRA fine-tuning. Practical recipes for inpainting,



super-resolution, and style transfer that ship to production. Hands-on: fine-tune SDXL with LoRA on a personal dataset and chain it with ControlNet for conditioned generation.

## Part IV — Multimodal, Video, and 3D

### Chapter 13

#### Vision-Language Models: The Multimodal Stack

Anatomy of a modern VLM: vision encoder + projector + LLM decoder. Covers LLaVA, Qwen2.5-VL, InternVL, Florence-2 / Florence-3, Gemma 3 Vision, and the closed frontier (GPT-4o, Claude Vision Sonnet 4.6/Opus 4.7, Gemini 2.5 Pro). When to fine-tune vs prompt vs use the API. Honest cost/latency/accuracy comparison table. Hands-on: build a chart-and-document Q&A system using the Claude Vision API + a local Qwen2.5-VL fallback.

### Chapter 14

#### Document AI and Visual Agents

The most underrated commercial CV application in 2026. Covers OCR (PaddleOCR, Tesseract, Donut), layout analysis (LayoutLMv3, Doctr), table extraction, structured-output VLMs, and the new visual agent loop: screenshot → reason → click. Walks through Claude Computer Use and OpenAI's vision agent pattern. Hands-on: build an invoice-extraction-to-database pipeline and a simple browser-automation agent driven by vision.

### Chapter 15

#### Video Understanding: Action, Tracking, and VideoMAE

Temporal modeling: optical flow (Lucas-Kanade, RAFT), 3D CNNs (I3D, SlowFast), VideoMAE, video-language models. Tracking pipeline: SORT → DeepSORT → ByteTrack → BoT-SORT → SAM 2 for video. Action recognition, temporal action detection, and video-text retrieval. Hands-on: build a sports-analytics pipeline that detects, tracks, and counts players' actions in a clip.

### Chapter 16

#### 3D Vision: Depth, NeRF, and Gaussian Splatting

Monocular depth (MiDaS, Depth Anything v2, Marigold), point clouds (PointNet, PointTransformer), structure-from-motion (COLMAP), NeRF and its descendants (Instant-NGP, Mip-NeRF), and 3D Gaussian Splatting — the 2024-2026 dominant scene representation for real-time photoreal rendering. Bridges to SLAM and AR/VR. Hands-on: capture a scene with a phone, train a 3DGS model in Nerfstudio, and render a fly-through.

# Part V — Frontiers and Production

## Chapter 17

### Vision-Language-Action Models for Robotics and Embodied AI

The category that didn't exist in 2023 and is now the most exciting in CV. Covers the VLA paradigm (vision encoder + language model + action decoder), RT-1/RT-2, OpenVLA,  $\pi 0$  (Pi-Zero, diffusion-based continuous control), Helix (Figure AI humanoid), and the Open X-Embodiment dataset. Discusses the pose estimation stack (OpenPose, MediaPipe, SMPL) as the bridge from pixels to actions. Hands-on: run OpenVLA in simulation on a manipulation task and visualize its attention over the scene.

## Chapter 18

### Domain-Specific Vision: Medical, Autonomous, Geospatial, Agricultural

How CV is applied where it actually generates revenue. Medical imaging (MONAI, nnU-Net, SAM-Med, regulatory considerations), autonomous driving (BEV perception, BEVFormer, lane detection, sensor fusion, nuScenes), satellite/geospatial (Segment Geospatial, foundation models for Earth observation), and agriculture (DINOv3 + YOLO26 for weed detection — a real 2026 paper). Hands-on: train a domain model in MONAI on a public chest-X-ray dataset and a BEV perception toy example.

## Chapter 19

### Optimization and Edge Deployment

Making models run fast on real hardware. Quantization (PTQ, QAT, INT8, FP16, FP8), pruning (structured/unstructured), knowledge distillation, ONNX export, runtime engines (ONNX Runtime, TensorRT, OpenVINO, Apple CoreML, TFLite, NVIDIA Triton), and edge platforms (Jetson Orin, Raspberry Pi 5 + Hailo, mobile NPUs). Honest benchmarks across the stack. Hands-on: take the YOLO26 from Chapter 7, quantize to INT8, export to TensorRT, and deploy on a Jetson Orin Nano with measured latency.

## Chapter 20

### MLOps, Responsible AI, and the Future of Computer Vision

The end-to-end production loop: data versioning (DVC), label management (Label Studio, Roboflow), active learning, drift detection, model monitoring, A/B testing for vision, CI/CD with MLflow/W&B.; Responsible AI: bias auditing, explainability (Grad-CAM, SHAP, Captum), privacy (federated learning, differential privacy for vision), and legal frontiers (EU AI Act, content provenance / C2PA). Closes with a forward-looking section: where CV is heading by 2028 (unified VLA agents, neural rendering as default, on-device foundation models). Capstone: ship the running case-study pipeline end-to-end as a monitored production service.

## 2-Month Writing Schedule (8 Weeks)

Assumptions: ~2 hours of focused writing/day ( $\approx$ 1,500 words on a good day) + 1 hour code/figures, 6 days/week. Each chapter  $\approx$  3 working days. Parallelism (researching one chapter while drafting the previous) compresses 60 working days into 8 calendar weeks. See **WRITING\_PLAN.xlsx** for the day-by-day grid.

| Week                | Focus                                                                                             |
|---------------------|---------------------------------------------------------------------------------------------------|
| Week 0 (Pre-launch) | Set up GitHub repo, pick running case-study dataset, draft chapter template, block writing hours. |
| Week 1              | Ch 1 (Landscape) → Ch 2 (Pixel ops) → outline Ch 3.                                               |
| Week 2              | Ch 3 (Classical features) → Ch 4 (Camera geometry, stereo).                                       |
| Week 3              | Ch 5 (CNNs ResNet-18) → Ch 6 (Training tricks).                                                   |
| Week 4              | Ch 7 (YOLO26 + RT-DETR) → Ch 8 (SAM 3 + Grounding DINO). Mid-book consistency pass.               |
| Week 5              | Ch 9 (ViTs) → Ch 10 (DINOv3, frozen-backbone notebook).                                           |
| Week 6              | Ch 11 (CLIP + FAISS) → Ch 12 (Diffusion) → Ch 13 (VLMs + Claude Vision).                          |
| Week 7              | Ch 14 (Document AI / agents) → Ch 15 (Video) → Ch 16 (3DGS) → start Ch 17.                        |
| Week 8              | Ch 17 (VLA) → Ch 18 (Domains) → Ch 19 (Edge) → Ch 20 (MLOps + Future). Cross-chapter pass.        |
| Week 9 (buffer)     | Full read-through, technical reviewer feedback, polish figures, write Preface.                    |

## Open Decisions

### Single framework or both?

PyTorch-only is faster to write and matches modern research; including TensorFlow/Keras doubles code maintenance but widens audience. Recommended: PyTorch-only with one TF appendix.

### Math depth?

"Understand enough to debug" (recommended, practitioner audience) vs "rigorous derivations" (academic audience). Pick one and hold the line.

### Running case-study dataset?

Recommend pairing MS-COCO (general) with a domain set you already know (medical, agriculture from EY, frog SDM) so domain examples come from real work.